🌍 Air Quality Data Analysis & Prediction

Problem:
Air-quality and environmental conditions fluctuate significantly over time, making it difficult to understand patterns and anticipate future conditions.

Approach:
I analyzed **9,357 hourly air-quality observations** using **Python/Pandas and Power BI**. The analysis explored pollutant concentrations, temperature, humidity, absolute humidity, sensor readings, and forecast behavior.

Key Analysis:

- Cleaned and analyzed environmental sensor data.
- Studied relationships between **RH and air-quality variables**.
- Compared **actual RH with forecasted values**.
- Visualized forecast ranges using **upper and lower uncertainty bounds**.
- Built an interactive Power BI dashboard for quick monitoring.

Key Findings:

- Actual RH averaged approximately **39.49%** in the dashboard's displayed KPI context, while the forecast averaged around **49.23%**.
- Forecast values ranged from approximately **20.17 to 74.78**.
- The forecast uncertainty extended from approximately **4.07 to 90.94**.
- In the cleaned dataset, actual RH and forecast showed a correlation of approximately **0.71**, indicating a meaningful relationship.
- The data contains substantial missing-value encoding (`-200`) in several sensor columns, highlighting the importance of data cleaning before modeling.

Outcome:
The project transforms raw environmental sensor data into an easy-to-understand **forecasting and monitoring dashboard**, helping users identify trends, compare predictions with observations, and understand prediction uncertainty.